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SH17: A dataset for human safety and personal protective equipment detection in manufacturing industry

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ABSTRACT

Workplace accidents continue to pose significant human safety risks, particularly in the construction and manufacturing industries. The necessity for effective Personal Protective Equipment (PPE) compliance has become increasingly paramount. We focus on developing non-invasive techniques based on the Object Detection (OD) and Convolutional Neural Network (CNN). The aim is to detect and verify the proper use of various types of PPE such as helmets, safety glasses, masks, and protective clothing. This study proposes the SH17 Dataset, consisting of 8,099 annotated images containing 75,994 instances of 17 classes collected from diverse industrial environments, to train and validate the OD models. We have trained state-of-the-art OD models for benchmarking, and initial results demonstrate promising accuracy levels with You Only Look Once (YOLO)v9-e model variant exceeding 70.9% in PPE detection. The validation of the model across cross-domain datasets indicates that integrating these technologies can substantially enhance safety management systems. This approach offers a scalable and efficient solution for industries seeking to comply with human safety regulations while safeguarding their workforce. The dataset is available at https://github.com/ahmadmughees/sh17dataset.

1. Introduction

Occupational Safety and Health (OSH) is a vital multidisciplinary domain focused on ensuring safe working environments and mitigating risks to worker health. It encompasses the prevention of work-related injuries, accidents, and illnesses by systematically managing workplace hazards and promoting health-conscious practices. In industrial manufacturing, PPE plays a pivotal role in protecting workers from various occupational hazards. The importance of PPE lies in its ability to significantly reduce the risks of injuries and illnesses resulting from exposure to hazardous conditions such as toxic chemicals, extreme temperatures, and mechanical dangers.

Extensive research has demonstrated that PPE serves as an essential last line of defense in safeguarding workers when other safety measures are inadequate [1,2]. Manufacturing industries have made considerable efforts to ensure workforce safety by mandating the use of PPE in their facilities, aiming to minimize injury-related incidents that can lead to substantial financial losses. Initially, human inspectors were tasked with monitoring PPE compliance at key points throughout the facility. However, this responsibility has increasingly been augmented by Internet of Thing (IoT) systems, which employ sensor-based methodologies to automate the verification of PPE compliance [3,4].

The advent of Machine Learning (ML) and Computer Vision (CV) systems has prompted a shift towards the development of fully non-invasive solutions for PPE compliance monitoring in manufacturing environments [5,6]. These advancements underscore the potential for more efficient and precise safety monitoring systems. In accordance with the Occupational Safety and Health Administration (OSHA) guidelines [7], Isalovic et al. [8] classified PPE into five categories based on the body regions they are designed to protect: (1) Head, (2) Upper Body, (3) Hands, (4) Feet, and (5) Whole Body. Typical examples of PPE include helmets, safety glasses, face shields, earmuffs, safety vests, gloves, and safety shoes, each tailored to mitigate specific hazards.

While proper selection, usage, and maintenance of PPE are critical for ensuring worker safety, recent studies [9–12] have increasingly utilized ML-based Object Detection (OD) models to detect PPE items across various environments, from construction sites to chemical plants. However, these studies focus on detecting specific PPE items, such as helmets or vests, within controlled environments [5]. Moreover, many of these approaches depend on proprietary or small-scale datasets, which restricts their generalizability and limits their broader applicability.

To overcome these limitations, this study expands the application of OD-based methods for PPE compliance by introducing a large-scale, high-quality, open-source dataset. Our dataset consists of 8099

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images and 75,994 labeled instances, serving as a valuable resource for advancing research in this domain. Furthermore, we benchmark the dataset against existing datasets and relevant studies, and we release open-source model weights for PPE detection, enabling validation and fostering further research within the community.

The main contributions of this work can be listed as follows:

- Curation of a high-quality, large-scale dataset, collected from the internet under a permissive license and open-sourced for research purposes. The dataset includes images from diverse global sources, ensuring inclusivity without location or gender bias.
- Extensive training of state-of-the-art OD models, with publicly available weights provided for community use.
- Rigorous evaluation of the trained models across different datasets to assess their efficacy on out-of-distribution data.

The remainder of this paper is structured as follows: a literature review and background are provided in Section 2. The dataset is introduced in Section 3. The insights from the experiments and results are provided in Section 4. Finally, Section 5 offers concluding remarks and future directions.

2. Literature review

Human safety inspectors remain the most prevalent and straightforward means of ensuring PPE compliance in the workplace. However, this approach is often associated with significant costs and labor intensity, as well as a substantial margin for error. Previously, the widespread adoption of IoT devices in industrial manufacturing introduced sensors designed to monitor PPE compliance. Nevertheless, this method is also characterized by high costs and sensitivity concerns [3,4]. Recently, CV methods have emerged as a non-invasive alternative, offering more economical and effective solutions compared to traditional sensors [8,13,14], and have gained acceptance in both construction and manufacturing sectors. These methods utilize annotated data to train ML models for the detection of required objects.

In this section, we discuss the existing datasets for PPE detection and the OD methods employed in the literature.

2.1. Existing datasets

Most existing datasets for PPE detection are tailored towards hardhat (often called helmets) detection in construction engineering.

2.1.1. Hardhat Wearing Detection (GDUT-HWD) dataset

The GDUT-HWD dataset [15] comprises hardhat images sourced from various internet platforms, featuring five distinct colors, each labeled individually. This dataset encompasses a total of 3174 images and 18,893 object instances of varying sizes. Notably, the publicly available version, which was compiled through crowd-sourcing,[1] contains numerous advertisement images that exhibit non-relevant backgrounds.

2.1.2. Safety Helmet Wearing (SHW) dataset

The SHW dataset [16] represents a comprehensive public resource[2] designed for the detection of both safety helmets and human heads. It comprises a total of 7581 images, encompassing 9044 instances of individuals wearing safety helmets and 111,514 instances of individuals not wearing safety helmets. The data was primarily sourced from search engines, with a portion derived from the SCUT-HEAD dataset [9], followed by meticulous manual annotation.

2.1.3. Color Helmet and Vest (CHV) dataset

The CHV dataset [17] is publicly accessible[3] and comprises 1330 high-quality images categorized into six distinct classes: persons, vests, and colored helmets in four different hues. The authors meticulously selected images from the GDUT-HWD [15] and SHW [16] datasets based on strict criteria, which included the relevance of the construction background, the gestures of the workers, and the angles and distances of the objects from the camera.

2.1.4. SHEL5K dataset

The SHEL5K dataset [10] is an improved version of Safety Helmet Detection (SHD) dataset[4] [18] consisting of 5000 images with 75,570 instances in 6 classes, including head, helmet, face, and person with and without a helmet. The dataset is publicly available.[5]

2.1.5. Pictor-PPE dataset

The Pictor-PPE dataset [5] is one of the earliest datasets collected for the PPE detection. It contains 3 classes, Hat, Vest, and Worker, and contains 774 crowd-sourced and 698 web-mined images. Its publicly available version[6] only contains 784 images available for scientific purposes from construction sites with a total of 2496 worker instances.

2.1.6. TCRSF dataset

The TCRSF dataset [6] is a closed-source dataset[7] and is derived from video feeds captured from various viewpoints within chemical plants. This dataset offers more realistic scenarios against complex backgrounds. It comprises 50,558 labeled instances across 12,373 images, categorized into seven classifications, including helmet, safety clothing, head, and others.[8]

All the aforementioned datasets, summarized in Table 1, have been compiled specifically to address the safety requirements of the construction industry. However, the broader context of human safety within general manufacturing environments remains inadequately addressed. In this study, we have examined this holistic perspective by collecting data from various environments.

2.2. YOLO models

You Only Look Once (YOLO) is a series of popular real-time object detection models that has been through several iterations from YOLOv1 to YOLOv8 [19–25]. The latest version, YOLOv8, builds upon its predecessors by incorporating state-of-the-art techniques and innovations to improve accuracy, speed, and adaptability.

YOLOv8 [23] follows a one-stage OD approach, where the input image undergoes a single pass through the network for bounding box prediction and classification. The architecture consists of three main components: a backbone network, a neck, and a prediction head following Redmon et al. [21] and Bochkovskiy et al. [22]. While the previous variants used the same head for objectness, classification, and regression tasks, Jocher et al. proposed the decoupled head along with Complete Intersection Over Union (CIoU) loss [26] and Distribution Focal Loss (DFL) [27], especially for the calculation of bounding box loss.

The major drawback of YOLOv8 was reliance on DarkNet-53[21]. It limited the ability to capture fine-grained features, especially for small and occluded objects. Wang et al. proposed YOLOv9 [28] to address this by introducing Programmable Gradient Information (PGI), a novel

[1] https://github.com/wujixiu/helmet-detection
[2] https://github.com/njvisionpower/Safety-Helmet-Wearing-Dataset
[3] https://github.com/ZijianWang-ZW/PPE_detection
[4] https://www.kaggle.com/datasets/andrewmvd/hard-hat-detection
[5] https://data.mendeley.com/datasets/9rcv8mm682/4
[6] https://github.com/ciber-lab/pictor-ppe
[7] The authors announced in [6] that the dataset will be publicly released at https://github.com/sofffty/TCRSF. However, it has not been released as of April 19, 2024.
[8] Information mentioned in their paper [6].

Table 1
Existing datasets for helmet and PPE detection.

Dataset	Classes	Images	Instances	Available	Paper
Pictor-PPE	3	784	–	✓	Nath et al. [5]
SHW	1	7,581	120,558	✓	njvisionpower [16]
CHV	6	1,330	–	✓	Wang et al. [17]
TCRSF	7	12,373	50,558	×	Yu et al. [6]
GDUT-HWD	5	3,174	18,893	✓	Wu et al. [15]
SHD	3	5,000	–	✓	–
SHEL5K	5	5,000	75,570	✓	Otgonbold et al. [10]
SH17 (proposed)	17	8,099	75,994	✓	–

concept that ensures the retention of critical information throughout the detection process. PGI integrates a reversible branch that works alongside the main network, preserving essential features and improving training outcomes without additional computational costs. Additionally, they proposed a Generalized Efficient Long-Range Attention Network (GELAN) block that optimizes the balance between parameter count, complexity, accuracy, and inference speed, enabling users to select the best computational blocks for various devices.

The recently proposed YOLOv10 [29] further builds upon predecessors and improves the inference speed by eliminating the non-max-suppression post-processing step which only picks the most probable bounding box out of 1000s of boxes produced by OD model. The authors proposed a consistent dual assignment strategy where a one-to-one assignment head is introduced in addition to a one-to-many head [30] with an identical structure. Both heads are jointly optimized during the training process by leveraging rich features; however, only a single head is employed during inference to enhance inference speed. Additionally, the authors reduced the number of trainable parameters by meticulously analyzing each trainable block and eliminating those that do not contribute significantly to overall efficiency. This optimization resulted in performance comparable to that of the YOLOv9-c variant, despite a reduction of 46% in trainable parameters. Section 2.3 discusses using these OD methods in the PPE detection for human safety.

2.3. Object Detection (OD) for PPE detection

Most prior studies concentrating on the utilization of OD for PPE detection have predominantly focused on applications within construction engineering, specifically aimed at verifying the use of hard hats and safety vests. Isailovic et al. [8] and Vukicevic et al. [12] proposed the PPE compliance using a two-stage approach by using a keypoint detector to detect regions and then passing these regions to the binary classification or OD model for further PPE detection. They used the images from Pictor-PPE [5] and hardhat dataset [31] to achieve this objective.

Wu et al. [32] used the Single Stage Detector (SSD) [33] architecture to identify hardhats of different colors on construction sites. They collected the novel GDUT-HWD dataset to train the model. Delhi et al. [34] and Tran et al. [35] employed YOLOv3 to detect hardhats and safety jackets in the construction environment and further used it to check for PPE compliance on various body parts [35] labeling safe and not-safe to each worker. They gathered the data from web sources. Otgonbold et al. [10] compared the performance of multiple OD models for detecting 6 different classes from person, helmet, head, and face on the SHEL5K dataset [10]. Nath et al. [5] used different combinations of OD approaches to address worker safety and PPE compliance. The authors initially detected the hard, vest, and worker and passed these to Machine Learning (ML) classifier, while in another approach, they trained the model to predict the state of the PPE. They passed the cropped RoI for a person to another CNN classifier for PPE classification. The Pictor-PPE dataset [5] is publicly available. Chen and Demachi [36] introduced a method using OpenPose [37] for body landmark detection and YOLOv3 [21] OD model for PPE detection. They analyzed the geometric relationships between the key points and detected PPE to assess compliance. They used the GDUT-HWD dataset for all their experiments. Zhafran et al. [11] explored the Fast R-CNN [38] architecture for checking masks, gloves, hardhats, and vests, noting a decrease in accuracy with greater distance and changes in lighting conditions using the manually collected laboratory data. Additionally, several studies have focused on detecting protective masks due to the Coronavirus Disease (COVID-19) pandemic. Loey et al. [39] used YOLOv2 while [40] combined SSD [33] with MobileNetV2 [41] for medical mask detection during COVID-19.

Ferdous and Ahsan [42] used YOLOX [43] OD model on the CHV dataset. Kim et al. [44] scrapped the internet to collect a dataset of 4844 images and manually annotated them into 3 classes, heavy, PPE, and worker to train YOLOv5 [23] and YOLOv8 [45]. Some other relevant works using OD for PPE detection include Lung and Wang [46], Xiang et al. [13], Di et al. [47], Han et al. [48], and Azizi et al. [49].

Some traditional approaches include combining the Histogram of Oriented Gradients with the Circle Hough Transform algorithms [50]. Li et al. [51] implemented a radiomics-based method for helmet detection, while Mneymneh et al. [52] introduced a motion detection-based system that subsequently identifies workers and hardhats. Balakrishnan et al. [53] designed a software system that includes an IoT module and Microsoft Azure's Custom Vision AI and Intelligent AI Services to detect safety glasses in lab settings. Amazon has also introduced the proprietary Amazon Rekognition PPE detection system [54].

It is evident that the majority of studies have been motivated by the specific needs of the construction industry, which continues to be one of the least digitized sectors and experiences a high incidence of fatal injuries. While these studies primarily concentrate on particular types of PPE, such as helmets, vests, and masks, the requirements of other industries remain largely underexplored. To the authors' knowledge, no comprehensive study has been conducted on PPE compliance within the manufacturing sector. Instead, existing research has targeted individual PPE types pertinent to specific industries. These studies typically employed a single architecture on custom or small-scale datasets, complicating direct comparisons due to the variations in the datasets utilized for training.

This study represents the first effort to comprehensively encompass and directly benchmark recent deep learning object detection architectures, offering an objective comparison facilitated by a newly developed dataset. The proposed object detection methodology aims to enhance compliance efficiency with multiple types of PPE across different body regions. Additionally, this research introduces a modular framework for PPE compliance that is applicable to various types of PPE and body parts.

3. SH17 dataset

Existing datasets for PPE detection frequently concentrate on specific scenarios, such as the detection of helmets, and may not adequately represent the diverse situations encountered in real-world industrial environments. In this study, we introduce the **S**afe **H**uman dataset, which comprises **17** distinct objects, hereafter referred to

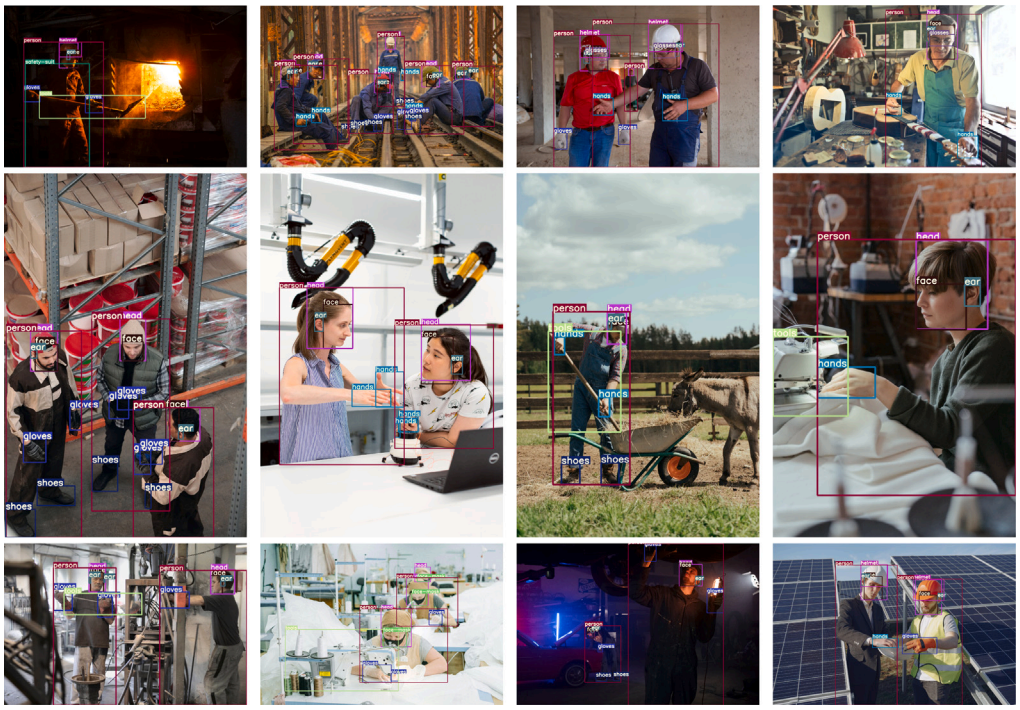


Fig. 1. Samples from the proposed **SH17 Dataset**. Best viewed online.

as the **SH17** dataset. We collected images from the Pexels[9] website, showcasing a variety of human activities across different industrial operations. Samples from the dataset are presented in Fig. 1. The subsequent sections will elaborate on the data collection process, annotation procedures, and additional details regarding the curation of this dataset.

3.1. Collection process

Many open-source datasets are derived from platforms such as Flickr[10] or compiled through web crawling using search engines such as Google or Bing. However, such data can be noisy and often necessitates substantial cleaning efforts, as images may be subject to varying licensing conflicts, leading to instances where studies do not publicly share their datasets. To facilitate a more streamlined process, we sourced images from Pexels, which provides clear usage rights for all its images. To identify relevant images, we employed multiple queries, including terms such as *manufacturing worker, industrial worker, human worker, and labor*. The tags associated with Pexels images were found to be reasonably accurate. After eliminating duplicate samples, we acquired approximately 11,000 images, of which around 26% were devoid of objects from our target classes. These empty images were excluded during the labeling process, resulting in a final dataset comprising 8099 images. The dataset demonstrates significant diversity, representing manufacturing environments globally, thereby minimizing potential regional or racial biases.

3.2. Annotation process

The annotation process engaged four human annotators, three initially completing the annotations. The team supervisor subsequently verified and corrected any discrepancies in the annotations. Finally, a graduate student conducted a thorough final verification and rectified any remaining mislabeling. All annotators employed DarkLabel[11] for data annotation, while the graduate student utilized LabelImg.[12] The choice of annotation tool was based on personal preference and did not affect the quality of the annotations.

3.3. Categories

The objects were categorized into 17 classes, each representing a specific body part and the requisite PPE items for safety. These classes were selected with the objective of enabling downstream applications to disregard irrelevant categories while still encompassing a wide range of scenarios in industrial manufacturing. The complete list of classes and their definitions is provided in Table 2. Additional tags associated with each class, designed to correspond with some of the classes in earlier datasets, were also included with each annotation and released as part of the extended version of the dataset. These tags are detailed in Table 2 and may be utilized by advanced users during training for downstream tasks. The metadata for each image sample was also extracted and provided with the dataset. The details are presented in Appendix A.

3.4. Data details

The original images extracted from Pexels have been provided in their native resolution, with a maximum image size of 8192 × 5462 and a minimum size of 1920 × 1002. The dataset comprises both landscape and portrait-style images, with each image containing an average of 9.38 instances. Annotations have been supplied in both Pascal VOC [55] and YOLO [19] formats to ensure compatibility with state-of-the-art models. Ears and earmuffs, which are considerably smaller objects in relation to a person, have also been labeled, thereby including objects of various sizes within the dataset. A total of 39,764 annotations contain less than 1% of the area, while 59,025 annotations account for less than 5% of the image area. Examples of all object sizes

[9] https://www.pexels.com/
[10] https://www.flickr.com/
[11] https://github.com/darkpgmr/DarkLabel
[12] https://github.com/HumanSignal/labelImg

Table 2
List of all the annotated classes and their counts.

ID	Name	Additional tags	Instances	Description
1	Person	male, female, children	13 802	Uses visible features for classification.
2	Head	–	11 985	Includes any view of the head: front, back, top or else.
3	Face	–	8950	Only classified as visible when the nose is visible.
4	Glasses	on, off, safety, vision	1945	Detection of safety glasses.
5	Face-mask-medical	on, off	669	Detection of medical face masks
6	Face-guard	on, off	134	Detection of whether faceguards
7	Ear		7730	Focused on the ears for safety equipment detection.
8	Earmuffs	on, off	318	Detection of earmuffs. Over-Ear-Headphones are labeled as earmufs.
9	Hands		15 850	Focus on the hands for safety equipment detection.
10	Gloves	on, off	2790	Detection of gloves.
11	Foot		796	Visible when there are no shoes, and each foot is annotated regardless of with or without socks.
12	Shoes	on, off, safety, other	4560	Shoes detection. Safety: Includes safety shoes and thick joggers. Others: slippers, sneakers, or other types of footwear.
13	Safety-vest	on, off	530	Detection of safety vests.
14	Tools	on, off	4647	Detection of tools being held; on means in hand. If the tool is present in the scene but not in hand, it's off. Pencils and laptops are not considered tools.
15	Helmet	on, off, white, red, black, yellow, blue	927	tags contain the color of a helmet as well.
16	Medical-suit	on, off	155	Detection of medical suits
17	Safety-suit	on, off	530	Detection of safety suits

Note: Tag off means the item is present in the scene but not worn by the person, while on means the item is present and worn by the person.

are illustrated in Fig. 1. The dataset includes the highest number of instances for hands, totaling 15,850, which constitutes 20.9% of the overall instances. The helmet class contains 927 instances, approximately 1.2% of the dataset, while the faceguard class comprises 134 instances, representing the lowest occurrence at 0.2%. A complete list of instances for each class is provided in Table 2. Additionally, Fig. 2 presents the distribution of all classes, their respective instances, and their percentage in the dataset, illustrating an imbalance in class distribution.

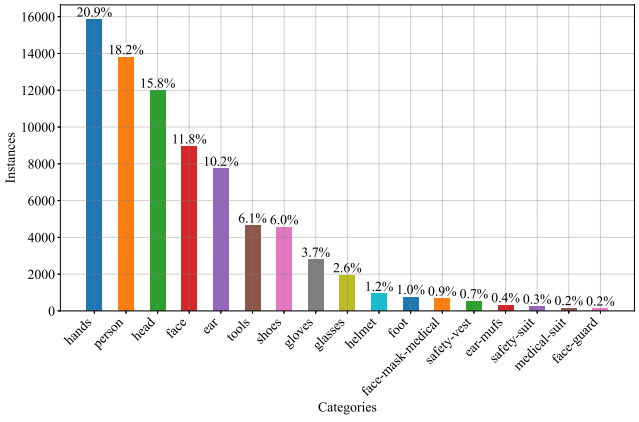


Fig. 2. Distribution of all class instances. The percentage of each class is shown at the top of each bar.

3.5. Evaluation metrics

Object detection models are evaluated using several metrics to assess their performance. Common metrics used by Microsoft Common Objects in Context (MS-COCO) dataset [56] and Pascal VOC dataset [55] include Precision (P), Recall (R), and Mean Average Precision (mAP).

Precision (P) measures the proportion of correctly identified objects among all objects detected by the model and is calculated as

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}$$

Recall (R), on the other hand, quantifies the ability of the model to detect all relevant objects in the dataset and is computed as

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}$$

Additionally, Mean Average Precision (mAP) is utilized to evaluate the detection accuracy across all classes. This metric is computed based on P and R for each class and subsequently averaged to produce an overall score. The Intersection over Union (IoU) metric is employed to assess the accuracy of object localization, calculating the overlap between the ground truth bounding boxes (b_g) and the model's predicted bounding boxes (b_{pred}) as follows:

$$\text{IoU} = \frac{\text{Area}(b_{pred} \cap b_g)}{\text{Area}(b_{pred} \cup b_g)}$$

where b_g represents the ground truth bounding box, and b_{pred} denotes the bounding box predicted by the OD model. The IoU threshold acts as a filter to remove FP bounding boxes with an IoU score below a certain threshold. This threshold determines the required accuracy for an object to be classified as detected or missed (e.g., IoU ≥ threshold). Different models may use various threshold values, such as 0.25, 0.5, or 0.75, during evaluation.

4. Experimentation and discussion

Multiple OD models were trained using various size variants of YOLOv8, YOLOv9, and YOLOv10 for benchmarking purposes. The dataset was divided into 80% for training and 20% for testing. Default

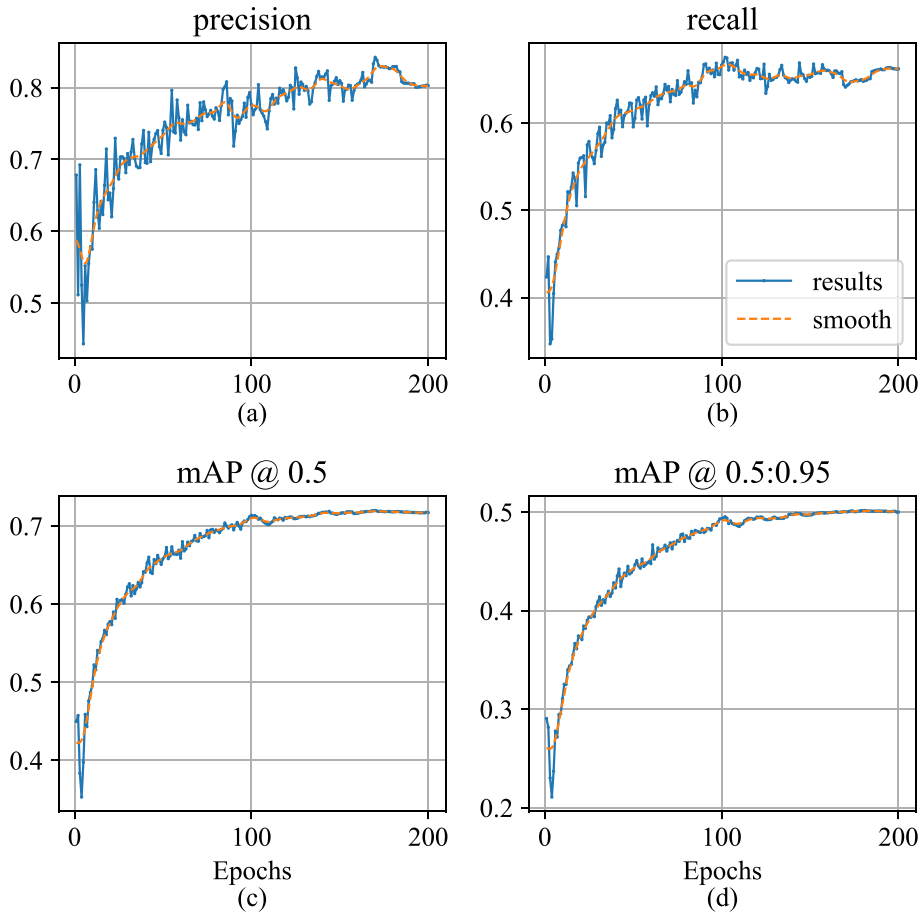


Fig. 3. Training metrics of YOLOv9-e model that performs best among all.

hyperparameters, as suggested by the respective authors in their open-source codebase,[13] were utilized. Transfer learning was employed to adapt a pre-trained model, initially trained on the MS-COCO dataset, which was then trained on the SH17 dataset for 200 epochs. The training hyperparameters were aligned with those presented in [45]. All models were trained on a fixed image size of 640 rather than the original high-quality images due to memory limitations imposed by two NVIDIA RTX GPUs, with batch sizes ranging from 128 for the nano models to 32 for larger scale models. A mosaic of four images, along with horizontal flipping, was employed for data augmentation, and Non-Max-Suppression (NMS) was applied for post-processing the outputs of the YOLOv8 and YOLOv9 model variants. The results of the model training were consolidated in Table 3, evaluated on a separate test set. As noted, YOLOv9-e demonstrated the best performance among all variants, achieving 70.9% mAP@50 and 48.7% mAP50-95. This model comprises 58.1M parameters, in contrast to the next best YOLOv8-x, which contains 68.2M parameters, representing an effective reduction of 15% in the number of parameters. Furthermore, YOLOv9-c exhibited comparable performance to YOLOv8-l while utilizing 32% fewer trainable parameters. Similarly, YOLOv10-x demonstrated comparable performance to YOLOv9-c, with a reduction of 15% in trainable parameters, significantly decreasing both training and inference times.

The training metrics for the YOLOv9-e model, identified as the best-performing model on the SH17 dataset, are presented in Fig. 3. It was observed that the training of the model plateaus at approximately 170 epochs, as the mAP50 does not show any improvement beyond this point. The class-wise accuracy of the YOLOv9-e model on the randomly sampled test set is detailed in Table 4. The hands class, which contains the most samples, demonstrates strong performance with an mAP of 89.8%. The samples for the tools and foot classes exhibit considerable variation in types, including safety shoes, slippers, joggers, and runners, while tools can encompass a wide range of items that a person may be utilizing. This variability has resulted in these classes exhibiting lower mAP scores.

These benchmarks further underscore the results reported in [28, 29,45], where YOLOv9 models demonstrate comparable performance with reduced parameters attributed to the implementation of PGI and GELAN. The YOLOv10 model encounters difficulties with small objects,

Table 3
Comparison of the models trained on SH17. Bold represents the best.

Model	Params	Images	Instances	*P*	*R*	mAP 50	mAP 50-95
Yolo-8-n	3.2	1620	15358	67.5	53.6	58.0	36.6
Yolo-8-s	11.2	1620	15358	81.5	55.7	63.7	41.7
Yolo-8-m	25.9	1620	15358	77.1	60.5	66.6	45.7
Yolo-8-l	43.7	1620	15358	76.7	62.9	68.0	47.0
Yolo-8-x	68.2	1620	15358	77.1	63.1	69.3	47.2
Yolo-9-t	2.0	1620	15358	75.0	52.6	58.5	37.5
Yolo-9-s	7.2	1620	15358	73.6	60.2	65.3	42.9
Yolo-9-m	20.1	1620	15358	77.4	62.0	68.6	46.5
Yolo-9-c	25.5	1620	15358	79.6	60.8	67.7	46.5
Yolo-9-e	58.1	1620	15358	**81.0**	**65.0**	**70.9**	**48.7**
Yolo-10-n	2.3	1620	15358	66.8	53.2	57.2	35.9
Yolo-10-s	7.2	1620	15358	75.8	57.0	62.7	40.9
Yolo-10-m	15.4	1620	15358	71.4	61.4	65.7	43.8
Yolo-10-b	19.1	1620	15358	77.7	59.1	65.8	45.1
Yolo-10-l	24.4	1620	15358	76.0	61.8	67.4	46.0
Yolo-10-x	29.5	1620	15358	76.8	62.8	67.8	46.7

[13] https://github.com/ultralytics/ultralytics

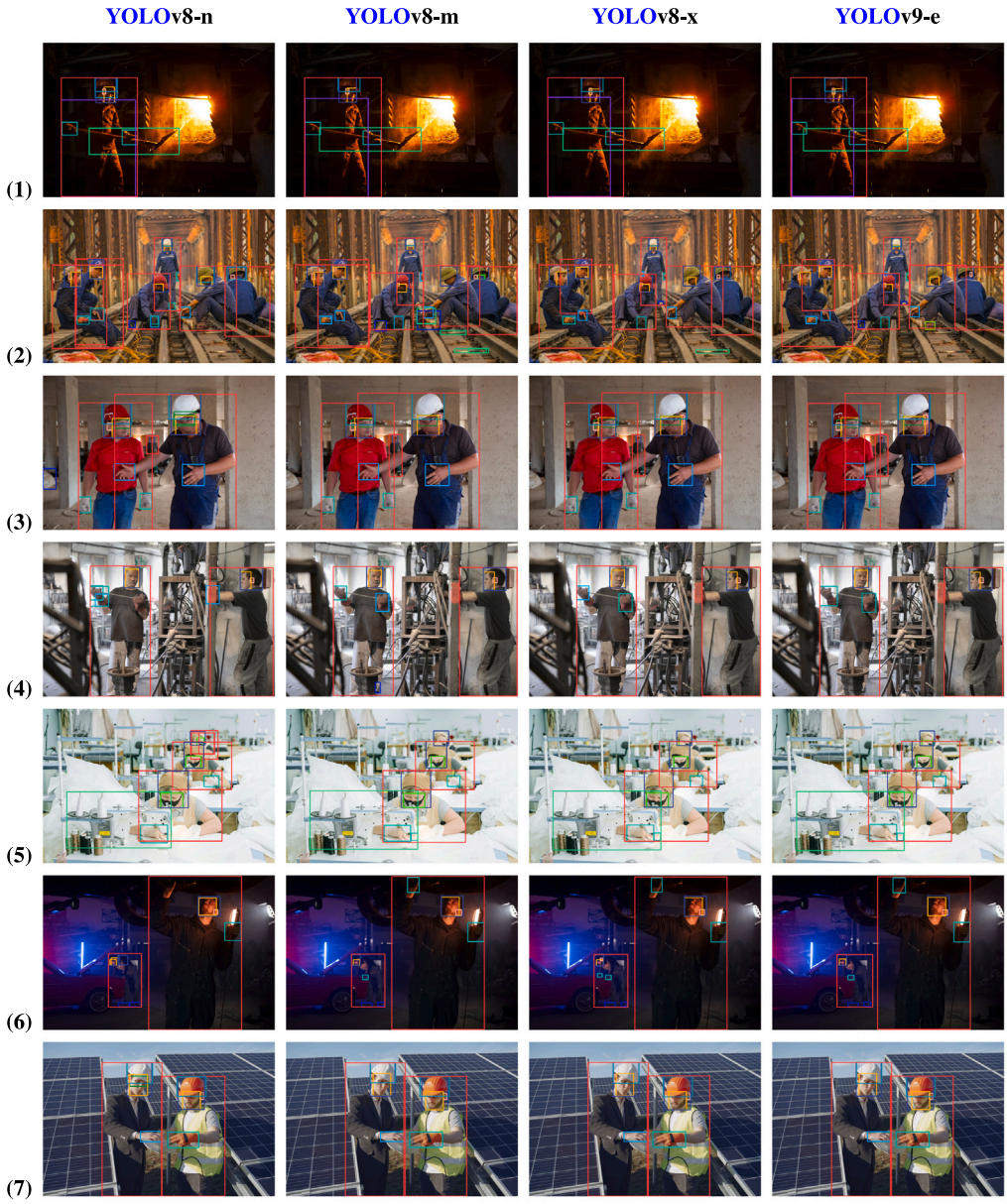


Fig. 4. Visual representation of the predictions made by different models; YOLOv8-n, YOLOv8-m, YOLOv8-x, YOLOv9-e. Best viewed online.

as the SH17 dataset comprises 52% of objects that occupy less than 1% of the area (as explained in Section 3.4). Fig. 4 illustrates some visual differences among the models: YOLOv8-n, YOLOv8-m, YOLOv8-x, and YOLOv9-e. In sample (2) of Fig. 4, both YOLOv8-m and YOLOv8-x made false positive predictions regarding the tool on the ground. For sample (3), the YOLOv8-n model produced a false positive prediction for the face class. In row 5, all models failed to detect the tool held by the worker. The original labeled samples are referenced in Fig. 1.

The efficacy of the model was verified using the SH17 dataset in conjunction with cross-domain datasets. The Pictor-PPE dataset [5] was selected to validate the performance of the YOLOv9-e model. This dataset comprises only three classes: worker, hat, and vest, which were mapped to the Person, Helmet, and Safety-vest classes of SH17, respectively. All publicly available images in the dataset were utilized for validation, resulting in a mAP of 58.8% for YOLOv9-e. This indicates that the trained models can be effectively employed for PPE compliance in manufacturing environments. Although the SH17 dataset contains the fewest samples for the safety-vest class, it still achieves a mAP of 35.5% at 50. The complete results are summarized in Table 5. Fig. 5 presents the confusion matrix for the entire Pictor-PPE dataset as evaluated by YOLOv9-e. It is evident that the person class is occasionally classified as background, while the helmet class is frequently misidentified as the head class. Visual results are also provided in Fig. 6.

5. Conclusion

Ensuring human safety in manufacturing environments is a paramount concern, particularly in maintaining PPE compliance. In this study, the SH17 dataset was introduced and open-sourced, comprising 8099 annotated images across 17 distinct classes that represent various PPE items and body parts. This dataset is freely available for research purposes and aims to advance the field of PPE detection in industrial settings.

To validate the utility of the SH17 dataset, several models were benchmarked, with YOLOv9-e achieving the highest performance. Additionally, the generalization capability of the trained model was tested using the Pictor-PPE dataset, with results indicating that the model is

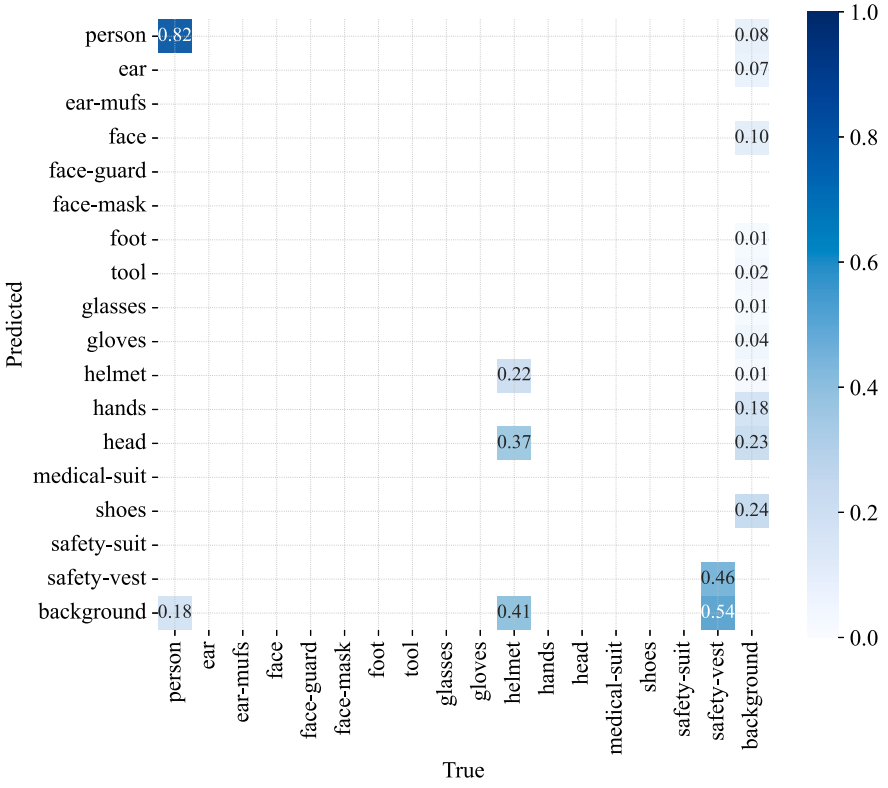


Fig. 5. Confusion Matrix of all the instances of Pictor-PEE dataset.



Fig. 6. Visual results of inference on Pictor-PPE dataset. Best viewed online.

well-suited for deployment in industrial environments to detect and ensure PPE compliance.

Looking ahead, future work could focus on enhancing the detection performance of minority classes within the dataset. This could be accomplished through the development of specialized models and the application of customized training methodologies. Such advancements are anticipated to further enhance the applicability of PPE detection systems in diverse and challenging industrial environments.

Table 4
Class-wise accuracy of YOLOv9-e model.

Class	Images	Instances	P	R	mAP 50	mAP 50-95
all	1620	15 358	81.0	65.0	70.9	48.7
hands	1284	3212	91.4	83.9	89.8	64.8
person	1515	2734	90.9	89.2	92.1	77.9
head	1314	2427	94.8	89.1	93.5	74.3
face	1155	1855	96.0	88.1	93.8	73.8
ear	987	1612	91.2	75.5	84.3	55.0
shoes	320	956	79.2	62.8	70.8	43.2
tool	455	923	67.4	39.1	43.2	27.6
gloves	254	529	81.6	58.9	66.5	43.5
glasses	323	398	87.4	72.6	76.4	46.9
helmet	93	154	81.3	67.8	77.0	57.6
face-mask	75	151	88.8	73.2	75.5	49.2
foot	64	149	51.7	22.1	29.3	14.0
safety-vest	45	97	66.4	55.0	57.7	38.1
ear-mufs	38	49	79.6	46.9	57.1	40.5
safety-suit	28	45	65.7	53.3	58.5	38.3
medical-suit	30	43	86.0	65.1	68.5	40.6
face-guard	23	24	76.8	62.5	71.7	42.8

Table 5
Class-wise accuracy of YOLOv9e model on the Pictor-PPE dataset [5].

Class	Images	Instances	P	R	mAP 50	mAP 50-95
all	654	3,477	73.5	50.3	58.9	37.6
person	654	2,080	83.6	80.7	85.5	61.6
helmet	451	1,369	94.6	23.7	55.5	30.0
safety-vest	12	28	42.4	46.4	35.5	21.2

Acronyms

AI	Artificial Intelligence.
CIoU	Complete Intersection Over Union.
CNN	Convolutional Neural Network.
COVID-19	Coronavirus Disease.
CV	Computer Vision.
DFL	Distribution Focal Loss.
FP	False Positive.
GELAN	Generalized Efficient Long-Range Attention Network.
IoT	Internet of Thing.
IoU	Intersection over Union.
mAP	Mean Average Precision.
ML	Machine Learning.
MS-COCO	Microsoft Common Objects in Context.
NMS	Non-Max-Suppression.
NSERC	Natural Sciences and Engineering Research Council of Canada.
OCI	Ontario Center of Innovation.
OD	Object Detection.
OSH	Occupational Safety and Health.
OSHA	Occupational Safety and Health Administration.
P	Precision.
PGI	Programmable Gradient Information.
PPE	Personal Protective Equipment.
R	Recall.
R-CNN	Region Proposal Convolutional Neural Network.
RoI	Region of Interest.
SSD	Single Stage Detector.
TP	True Positive.
YOLO	You Only Look Once.

Table A.1
Metadata format for the dataset.

Field	Description
Unique Identifier	Unique code for each image for identification.
Width and Height	Dimensions of the image, specified in pixels.
URL	Web address where the image can be accessed.
Photographer Name	Name of the individual who captured the image.
Photographer URL	Web address to the photographer's portfolio or profile.
Photographer ID	Unique identifier for the photographer, used for reference or database purposes.
Average Color	Average color in the hexadecimal code.
Source	Origin from which the image was obtained.
Liked	Indicates whether the image has been marked as liked or favored on the online platform, a boolean value.
Description	Summary of the image's content, context, or theme.

CRediT authorship contribution statement

Hafiz Mughees Ahmad: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Afshin Rahimi:** Writing – review & editing, Visualization, Supervision, Software, Resources, Project administration, Funding acquisition, Formal analysis.

Institutional review

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Ethics statements

Not applicable because this work does not involve the use of animal or human subjects.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. SH17 dataset

A.1. Meta data

The collected meta-data for each image has also been provided, which can be utilized to reconstruct the dataset from the source or can serve as supplementary data during model training. The list of objects and their definitions included in the meta-data is detailed in Table A.1. Additionally, scripts for constructing the data from the source are available in the SH17 dataset GitHub repository.

Table A.2
Quantitative analysis of the additional tags with Persons class.

	Male	Female	Children	Total
White	2,432	2,032	37	4,501
Black	1,098	776	7	1,881
Brown	577	218	12	807
Asian	963	1,272	52	2,287
Total	5,070	4,298	1,08	9,476

Table A.3
Sorted list of instances in each class in Train and Test Set.

Class	Train	Test
face-guard	110	24
medical-suit	114	43
safety-suit	195	45
ear-mufs	269	49
safety-vest	433	97
face-mask-medical	519	151
foot	610	149
helmet	773	154
glasses	1,547	398
gloves	2,261	529
shoes	3,604	956
tools	3,724	923
ear	6,118	1,612
face	7,095	1,855
head	9,558	2,427
person	11,068	2,734
hands	12,638	3,212

A.2. Person demographics

In addition to the bounding boxes representing the Persons class, additional tags based on visible features within the images have been included. While these tags are primarily derived from observable characteristics, some inaccuracies may be present, as visual features are not consistently reliable indicators. Nonetheless, these tags have been incorporated to identify and mitigate potential biases related to gender or ethnic representation. The tags encompass categories such as male, female, children, and various ethnic backgrounds, including Black, Brown, White, and Asian. A quantitative analysis of these tags is provided in Table A.2.

A.3. Train test split

In this study 80% of the data was allocated for training purposes, while 20% was designated for testing. A separate list of the training and test files is provided in the SH17 dataset GitHub repository. The complete list of instances in both the training and test sets is presented in Table A.3.

Appendix B. Supplementary materials

Supporting data is available at SH17 dataset GitHub repository.

Data availability

The data presented in this study are available in SH17 dataset GitHub repository at https://github.com/ahmadmughees/sh17dataset.

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